CS204: Algorithms & Data Structures Lab

Week # 08 (1 Questions, 111 Marks)
Lab session: AL1
Held on: 23-Sep-2024 (Mon)
Lab Timings: 14:00 to 17:00 Hours Pages: 3
Submission time: 16:45 Hrs, 23-Sep-2024
Instructors Dr. V. Vijaya Saradhi
Head TAs Adithya K Moorthy & Laxita Agrawal
Department of CSE, IIT Guwahati

Question 1: (111 points)

Linked Lists addition of two binary numbers linked lists.

Task 01 (5 marks) Declare a class `Node` with the following specification

1. Three private members
   (a) (1 mark) An `int` representing a bit in a binary number
   (b) (1 mark) An `int` representing position of the bit in the binary number
   (c) (1 mark) A pointer to `class Node`
2. (1 mark) A public constructor - `Node` which initializes the data members.
3. (1 mark) Make `class LinkedList` a friend

Task 02 (60 marks) Declare a class `LinkedList` with the following specification.

1. (1 mark) Private member - `A pointer to class Node`
2. The following public member functions
   (a) (10 marks) A function named `add_node` with the following specifications
      i. **input argument**: an `int`
      ii. **output**: `void`
      iii. **Functionality**: This function should add a new `Node` to the linked list at the end.
   (b) (5 marks) A function named `add_node_front` with the following specifications
      i. **input argument**: an `int`
      ii. **output**: `void`
      iii. **Functionality**: This function should add a new `Node` to the linked list at the front of the linked list
   (c) (5 marks) An overloaded << `operator` with the following specification
      i. **Input argument 1**: `ostream` reference
      ii. **Input argument 2**: `const LinkedList` reference
      iii. **Returns**: `ostream` reference
      iv. **Functionality**: Print, to standard terminal, the linked list.
   (d) (25 marks) An overloaded + `operator` with the following specification

i. **Input argument 1**: `LinkedList` object

ii. **Returns**: `LinkedList` object

iii. **Functionality**: Adds the two linked lists and returns a new linked list.

(e) (5 marks) `get_node` member function with the following specifications

i. **input argument**: an `int` refer to the position of the node

ii. **output**: `int`

iii. **Functionality**: This function should return the bit value located at the given position.

(f) (4 marks) A destructor which frees the created memory for the linked list.

(g) (5 marks) A function `decimal` with the following specifications

i. **Input argument 1**: `void`

ii. **Returns**: `int`

iii. **Functionality**: Returns the decimal number corresponding the binary number of the linked list

Input file description

1. Each input file contains two lines
2. First line corresponds to first binary number
3. Second line corresponds to the second binary number
4. Each line has one or more numbers from the set {0, 1}
5. End of each line is denoted with -1
6. Number of bits in the first binary number is always equal to or greater than number of bits in the second binary number.

Task 03 (18 marks) Perform the following in the main function

1. (2 marks) Read the elements of first line into one lined list
2. (2 marks) Read the elements of second line into another linked list

3. (1 mark) Print the first linked list
4. (1 marks) print the decimal number in the first linked list

5. (1 mark) Print the second linked list
6. (1 mark) Print the decimal number in the second linked list

7. (8 marks) perform binary number addition using the above two linked lists and store it in a third linked list.
8. (1 mark) Print the third linked list
9. (1 mark) Print the decimal number in the third linked list

Task 04 (28 marks) Perform the following in the main function using STL

1. (2 marks) Read the elements of first line into `forward_list` of STL

2. (2 marks) Read the elements of second line into another `forward_list` of STL
3. (1 mark) Print the first `forward_list`
4. (1 mark) Print the decimal value of the first `forward_list`
5. (1 mark) Print the second `forward_list`
6. (1 mark) Print the decimal value of the second `forward_list`
7. (18 marks) perform binary number addition using these two `forward_list`s. Store the result in a third `forward_list`
8. (1 mark) Print the resulting `forward_list`
9. (1 mark) Print the decimal value of the third `forward_list`